SMART INDIA HACKATHON 2026

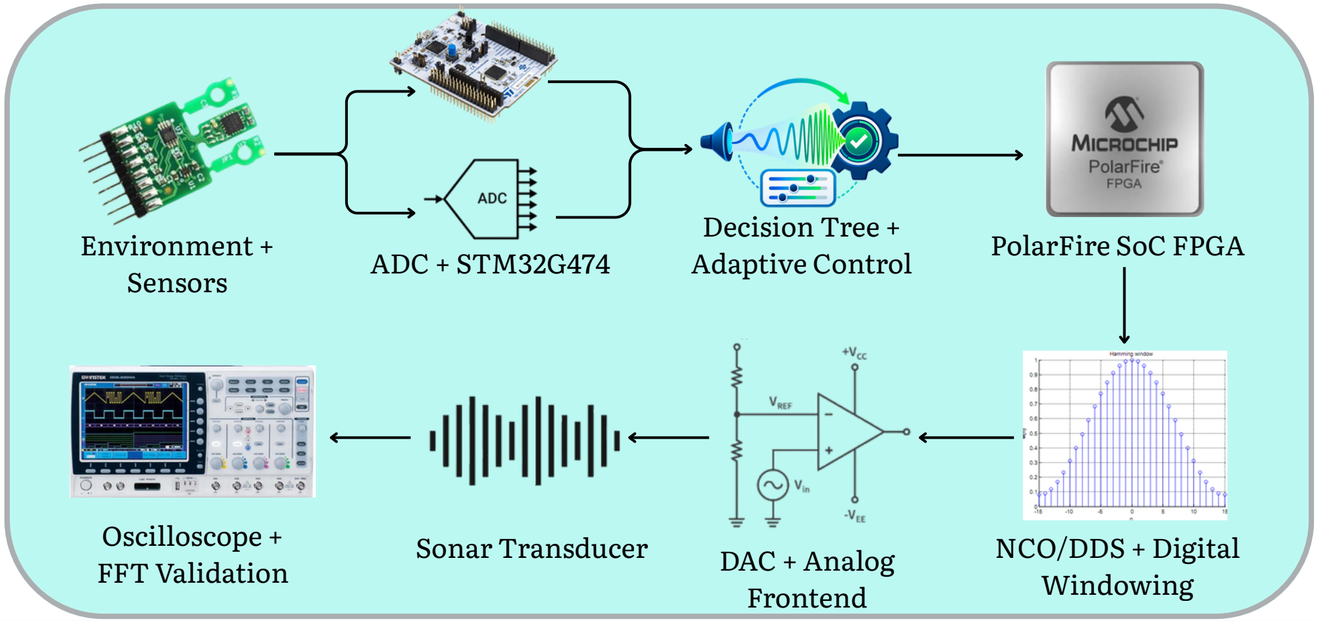
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- **Problem Statement ID** – SIH26058
- **Problem Statement Title** - Development of a Low-Power, Real-Time Adaptive Software-Defined Sonar Transmitter Payload for Autonomous Underwater Vehicles (AUVs)
- **Theme** - Robotics & Drone
- **PS Category** - Hardware
- **Team Name** - Chirp Craft



PROPOSED SOLUTION

- **AI/ML** using a **Decision Tree** classifies real-time temperature, turbidity, salinity and depth data into mixed underwater **environment fingerprints**.
- **Adaptive optimization** with **hysteresis** dynamically selects frequency, bandwidth, pulse duration and signal amplitude, **reducing transmit power** when lower strength is sufficient.
- **PolarFire SoC FPGA** using **NCO/DDS** and fixed-point DSP generates **LFM, geometric-sweep and phase-coded pulses** with **Hamming windowing** and deterministic **DMA streaming** to the **DAC**.
- **Hysteresis-based** adaptive gain control prevents rapid **amplitude switching** during **environmental transitions**, ensuring stable and efficient sonar transmission.



INNOVATION & UNIQUENESS

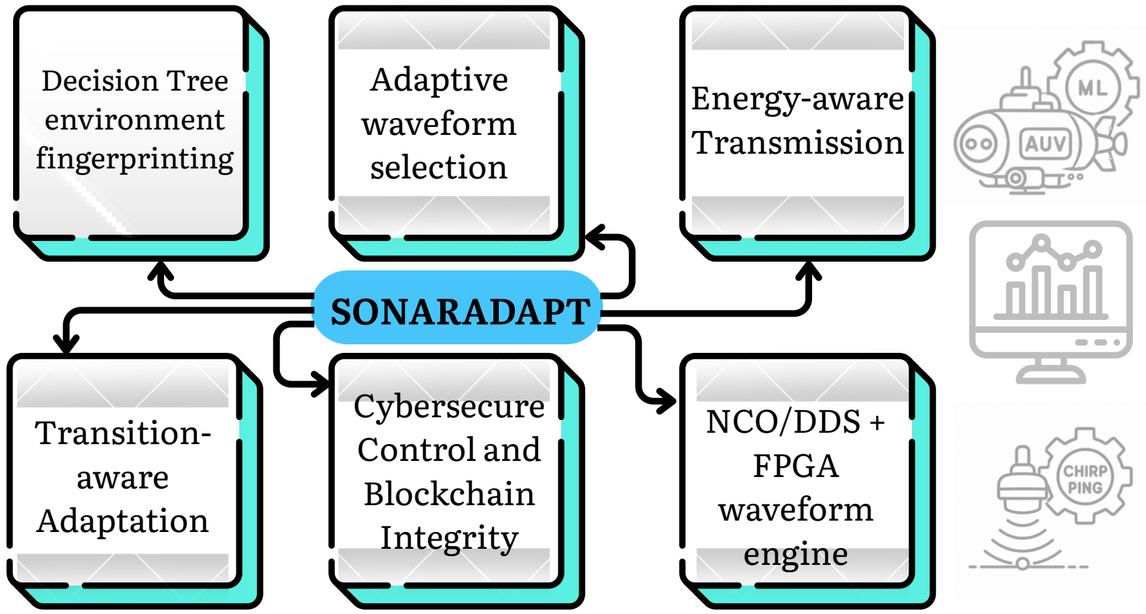
➤ ***Decision Tree–Based Environment Fingerprinting***
Fuses temperature, turbidity, salinity and depth to classify mixed conditions.

➤ ***Hysteresis-Based Adaptive Waveform Selection***
Dynamically tunes frequency, bandwidth, pulse duration and amplitude.

➤ ***FPGA NCO/DDS + Fixed-Point DSP Acceleration***
Enables deterministic LFM, sweep and phase-coded waveform generation.

➤ ***Transition-Aware Confidence Logic***
Prevents unstable waveform switching during changing environments.

➤ ***Energy-Aware Adaptive Transmission***
Reduces signal amplitude energy when lower transmit strength is sufficient.



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TECHNICAL APPROACH

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Real-Time ML-Driven Adaptive Sonar Transmitter with FPGA Waveform Synthesis

System Input Domain
(Environmental Parameters)

Environmental Sensors & Emulation:

- **TSYS01** Temperature (°C)
- **MS5837-30BA** Depth (m)
- **SEN0710** Turbidity
- **SEN0451** Salinity

(I²C) - Digital | (I²C) - Digital | RS-485/Modbus | Analog Volt(PPT)

Sensor Interface Hub

Converts real-world physical parameters to digital data (e.g., 12-bit, 10 kS/s)

Software Control Domain

STM32G474 Adaptive Control Engine

Environmental Intelligence Engine

UnderWater Environment FingerPrinting

Sensor Fusion → Decision Tree → State Determination → Clear Shallow Reef / Deep Open Water / Muddy Estuary / Transition Zone

Adaptive Waveform Optimization

Decision Matrix (e.g., If **Deep_Mud** Then **Frequency**=150 kHz, **BW**=20 kHz) → Optimal **Chirp** Parameters(**fc**,**BW**, **Pulse Duration**)

Energy-Aware Transmission Control

Available Power Budget (P_{budget}) + Required Signal Strength → Amplitude & Pulse Duration Optimization

Context-Aware Acoustic Command Orchestration

Dynamically maps environmental states to optimal sonar parameters

FPGA Waveform Command Interface

- Waveform Type
- Fc, Bandwidth
- Pulse Duration
- Amplitude
- Window Type
- Sample Rate

STM32 decides WHAT to transmit

PolarFire SoC FPGA Waveform Engine

Command Interface & Parameter Registers: Receives data from STM32

Waveform Generation Engines (Parallel):

LFM Chirp | Geometric Sweep | Phase coded Pulse

Waveform MUX

NCO / DDS Core: High Resolution Phase Accumulator

Digital Windowing: Hamming Windowing

Amplitude Scaling: Fixed-Point Multiplier

Pulse Controller: Length, Repetition, Timing

Buffering & Streaming: Ping-Pong BRAM Buffers → DMA / Streaming Controller

Hardware Timer-Triggered DMA : Deterministic clocking for real-time waveform streaming.

FPGA generates HOW to transmit

Analog Front End (AFE) Domain

Signal Generation & Conditioning:

DMA to DAC

Reconstruction **Low-Pass Filter (LPF)** (Active Sallen-Key)

Variable Gain Amplifier / Gain Control (Controlled by MCU)

Hardware Bandpass Filter (BPF)

(Passes 100-500 kHz)

$$f_c = \sqrt{f_L f_H}$$

Power Amplification & Emission Domain

Power Management & Emission:

Power Amplifier (PA) Stage (e.g., Class-D for high efficiency, 10W)

Sonar Projector / Transducer

Acoustic Ping to Underwater Environment

Output Testing & Validation Domain

Oscilloscope for Time-Domain Validation:

Time Domain LFM Chirp Waveform

Observe Pulse Shape, Duration, Amplitude

ADC-based Spectrum Analyzer for Frequency-Domain Validation:

PA Output Test Point → ADC → **MCU** (FFT) → Display

FFT

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-2\pi kn/N}$$

Hamming Window & Sample FFT / Time Domain Spectrum Display

SYSTEM PIPELINE & LOGIC SUMMARY

Environmental Inputs (Sensors/Potentiometers) → **Environmental Fingerprinting & Optimization** (MCU/FPGA) → **Real-time Waveform Generation** (MCU/FPGA + DMA) → **Real-time Waveform Generation** (MCU/FPGA + DMA) → **Analog Conversion & Signal Conditioning** (DAC + AFE)

Problem Solved: Efficient adaptive sonar transmissions in changing underwater environments

Key Feutures: Deterministic DMA Streaming, Adaptive Hysteresis, Multi-Sensor Fusion, Energy-Aware Amplitude Control.

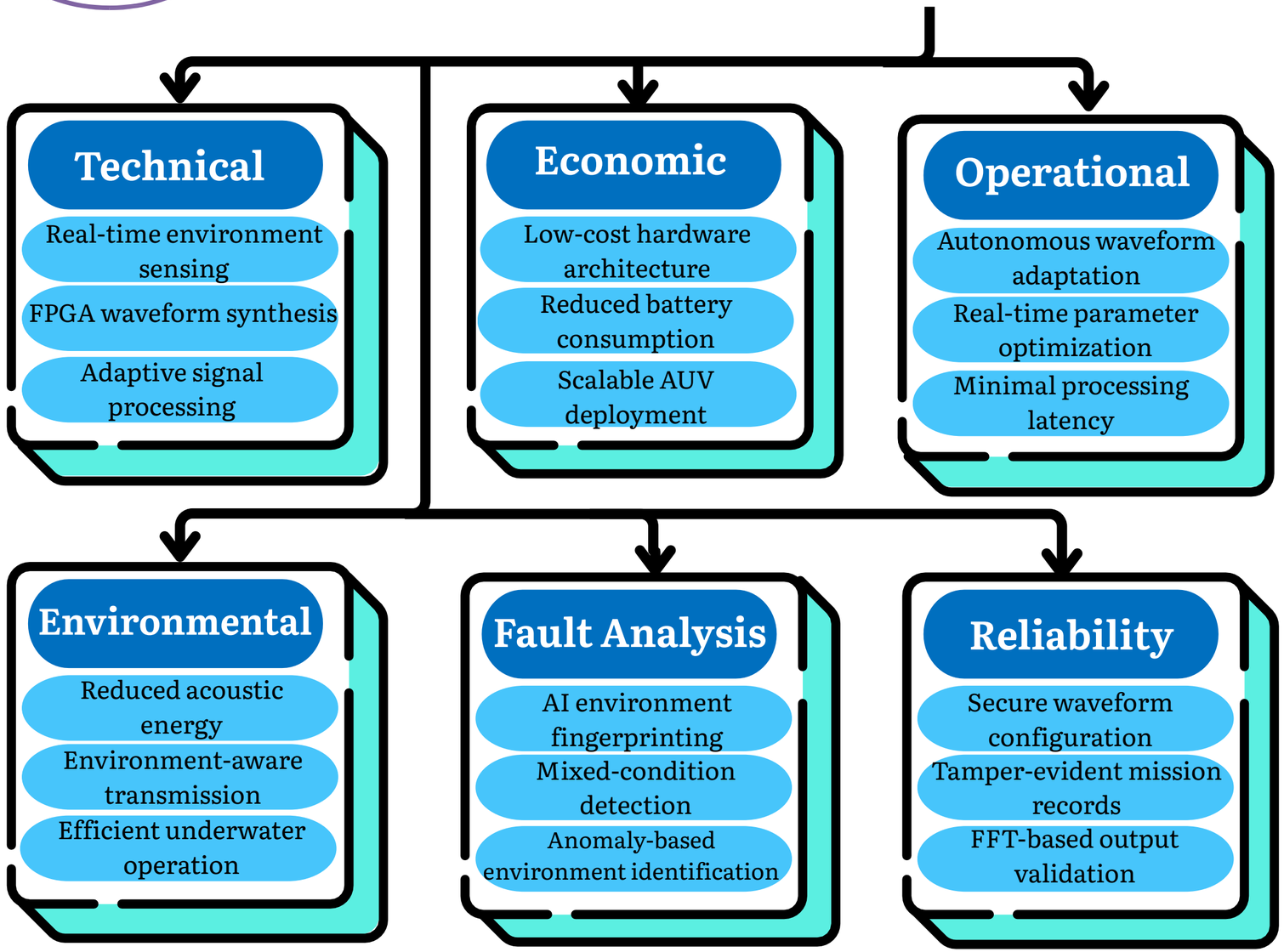
Design Parameters:

- Sensors: NTC Thermistor, PT100
- Memory Constraint: 1 KB BRAM/pulse
- Output Acoustic Pings: 10 W
- PA: Class-D
- Projector: 100-500 kHz Omnidirectional

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FEASIBILITY AND VIABILITY

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Solution Features and Scope Assessment

Real-time environmental sensing enables continuous adaptive sonar operation.	✓
AI-based fingerprinting identifies complex mixed underwater conditions accurately.	✓
Dynamic waveform selection balances resolution, range, and penetration.	✓
Energy-aware control reduces unnecessary transmission power consumption significantly.	✓
FPGA DDS generation provides precise low-latency waveform synthesis.	✓
Digital windowing reduces sidelobes and smooths pulse transitions.	✓
Analog filtering ensures clean, stable transducer drive signals.	✓
Cybersecurity protects firmware, models, and waveform configurations against tampering.	✓
Blockchain-backed records provide tamper-evident mission data integrity assurance.	✓
Oscilloscope and FFT verify transmitted waveform quality experimentally.	✓

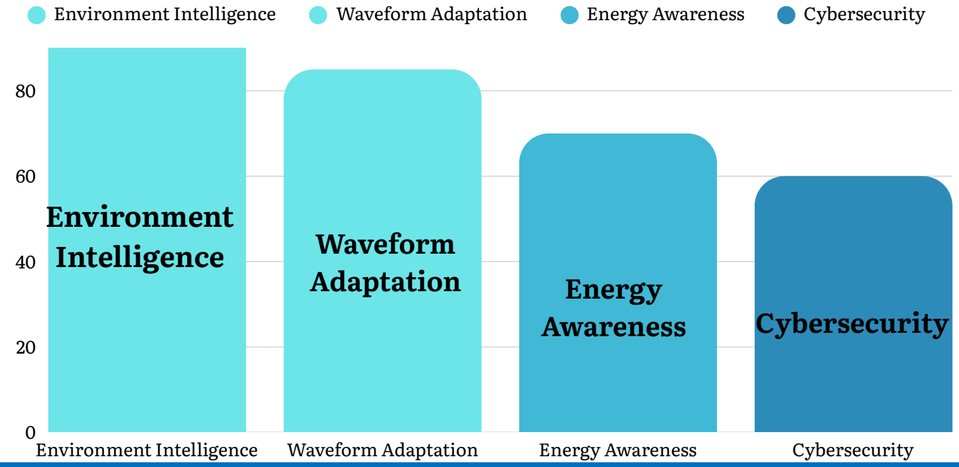
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IMPACT AND BENEFITS

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Impact Area	Description	SDG Connection
Adaptive Marine Mapping	Improved sonar adaptability across changing underwater environments.	SDG 14 (Life Below Water), SDG 9 (Industry, Innovation & Infrastructure)
Energy Efficiency	Reduced unnecessary transmission energy for battery-limited AUVs.	SDG 7 (Affordable & Clean Energy), SDG 12 (Responsible Consumption)
Intelligent Exploration	ML-based classification of complex underwater environmental conditions.	SDG 9 (Industry, Innovation & Infrastructure), SDG 14 (Life Below Water)
Reliable Autonomous Operation	Stable waveform adaptation during changing environmental conditions.	SDG 9 (Industry, Innovation & Infrastructure)
Cybersecure Marine Sensing	Protected firmware, configurations and mission integrity records.	SDG 9 (Industry, Innovation & Infrastructure), SDG 16 (Peace, Justice & Strong Institutions)

Metric	Target
Environment Adaptation	Dynamic
Waveform Selection	3 Modes
Parameter Optimization	4+ Parameters
Energy Management	Adaptive
Mission Integrity	100% Verifiable*



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RESEARCH AND REFERENCES

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Purpose: Optimizes sonar waveforms according to underwater conditions.
Source Content: Uses environmental information to optimize LFM waveform.
Justification: Supports our adaptive frequency and bandwidth selection approach.



Enhanced Target Detection Using a New Cognitive Sonar Waveform Design in Shallow Water

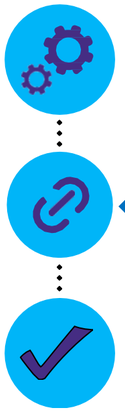
Application of Machine Learning Models in Water Quality Classification



Purpose: Adapts signal parameters using real-time underwater conditions.
Source Content: Uses Decision Tree and Random Forests for adaptive selection.
Justification: Supports our lightweight ML environment fingerprinting approach.

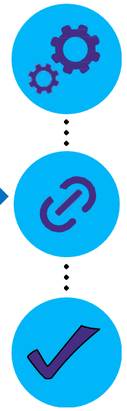
SONARADAPT

Purpose: Generates precise digital waveforms using FPGA hardware.
Source Content: Implements NCO/DDS using digital phase accumulation.
Justification: Supports our FPGA-based DDS/NCO waveform generation architecture.



NCO / DDS Periodic Waveform Generator for FPGA

Blockchain for Unmanned Underwater Drones



Purpose: Explores secure data management for underwater drones.
Source Content: Examines blockchain for security, storage and resource management.
Justification: Supports our blockchain-based mission-data integrity layer.

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TECHNICAL APPROACH

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System Input Domain
(Environmental Parameters)

Environmental Sensors & Emulation:

TSYS01
Temperature (°C)



MS5837-30BA
Depth (m)



SEN0710
Turbidity



SEN0451
Salinity



(I²C) - Digital | (I²C) - Digital | RS-485/Modbus | Analog Volt(PPT)

Sensor Interface Hub

Converts real-world physical parameters to digital data (e.g., 12-bit, 10 kS/s)

Software Control Domain
(MCU/FPGA-based Adaptive Sonar Controller)

STM32 Sonar Intelligence Engine

UnderWater Environment FingerPrinting

Sensor Fusion → Classification Algorithm → State Determination → Clear Shallow Reef / Deep Open Water / Muddy Estuary / Transition Zone

Adaptive Waveform Optimization

Decision Matrix (e.g., If **Deep_Mud** Then **Frequency**=150 kHz, **BW**=20 kHz) → Optimal **Chirp** Parameters(**fc**,**BW**, **Pulse Duration**)

Energy-Aware Transmission Control

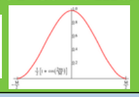
Available Power Budget (P_{budget}) + Required Signal Strength → Amplitude & Pulse Duration Optimization

Transition - Aware Adaption

Raw Sensor Filtering + Confidence Calculation → Hysteresis-based Switching Control → Stable Waveform Command

Waveform Generator & Windowing (Hamming):

$$w(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{M}\right)$$



$$x(t) = A\sin\left(2\pi\left(f_0 t + \frac{k}{2}t^2\right)\right)$$

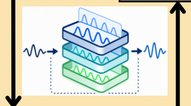
Hierarchical Memory & Data Straming Domain

Memory Management & DMA Controller:

DMA

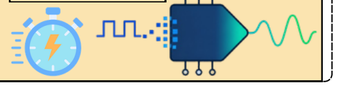
Waveform Buffers (e.g., Ping-Pong Buffers, 1024 Samples)

STM32 / FPGA BRAM Features (e.g., Feature Tiling)



Timer Triggering fs and pulse period Tping

DAC



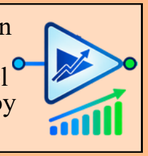
Analog Front End (AFE) Domain

Signal Generation & Conditioning:

DMA to DAC

Reconstruction **Low-Pass Filter (LPF)** (Active Sallen-Key)

Variable Gain Amplifier / Gain Control (Controlled by MCU)



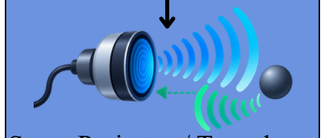
Hardware Bandpass Filter (BPF)
(Passes 100-500 kHz)

$$f_c = \sqrt{f_L f_H}$$

Power Amplification & Emission Domain

Power Management & Emission:

Power Amplifier (PA) Stage (e.g., Class-D for high efficiency, 10W)



Sonar Projector / Transducer



Acoustic Ping to Underwater Environment

Output Testing & Validation Domain

Oscilloscope for Time-Domain Validation:



Time Domain LFM Chirp Waveform

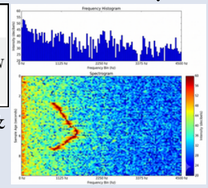
Observe Pulse Shape, Duration, Amplitude

ADC-based Spectrum Analyzer for Frequency-Domain Validation:

PA Output Test Point → ADC → **MCU** (FFT) → Display

FFT

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-2\pi k n/N}$$



Hamming Window & Sample FFT / Time Domain Spectrum Display

SYSTEM PIPELINE & LOGIC SUMMARY

Environmental Inputs (Sensors/Potentiometers) → **Environmental Fingerprinting & Optimization** (MCU/FPGA) → **Real-time Waveform Generation** (MCU/FPGA + DMA) → **Real-time Waveform Generation** (MCU/FPGA + DMA) → **Analog Conversion & Signal Conditioning** (DAC + AFE)

Problem Solved: Efficient adaptive sonar transmissions in changing underwater environments

Key Feutures: Deterministic DMA Streaming, Adaptive Hysteresis, Multi-Sensor Fusion, Energy-Aware Amplitude Control.

Design Parameters:

- Sensors: NTC Thermistor, PT100
- Memory Constraint: 1 KB BRAM/pulse
- Output Acoustic Pings: 10 W
- PA: Class-D
- Projector: 100-500 kHz Omnidirectional

Your Team Name

TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

1. System Input Domain
(Environmental Parameters)

Underwater Sensors & Interface:

- Temperature (NTC/PT100)
- Turbidity (NTU)
- Salinity (PPT)
- Depth (Pressure)

ADC & Signal Conditioning
(12-bit / 16-bit ADC)

Sensor Interface Hub
(I2C / SPI / UART)

Digitizes and conditions real-world environmental parameters for intelligent decision making.

2. STM32G474 Adaptive Control Engine

Environmental Intelligence

Preprocessing & Filtering → Feature Extraction → ML Classifier (Decision Tree) → Environment Fingerprint

Adaptive Optimization

Transition-Aware Logic (Hysteresis) → Waveform Selection → Parameter Optimization (Fc, BW, Pulse, Amp) → Energy-Aware Control

System Management

- Power Budget Monitoring
- Mode & Configuration Control
- Security & Access Control
- Health Monitoring & Logging

Output to FPGA

Waveform Command Packet (SPI / UART Interface)

- Waveform Type
- Fc, Bandwidth
- Pulse Duration
- Amplitude
- Window Type
- Sample Rate

STM32 decides WHAT to transmit

3. PolarFire SoC FPGA Real-Time Waveform Engine

Command Interface & Parameter Registers
(From STM32)

Waveform Generation Engines (Parallel)

- LFM Chirp Engine
- Geometric Sweep Engine
- Phase-Coded Pulse Engine

Waveform MUX

NCO / DDS Core
(High Resolution Phase Accumulator)

Digital Windowing
(Hamming / Hann / Blackman)

Amplitude Scaling
(Fixed-Point Multiplier)

Pulse Controller
(Length, Repetition, Timing)

Buffering & Streaming

Ping-Pong BRAM Buffers → DMA / Streaming Controller

Deterministic Sample Clock Generator

FPGA generates HOW to transmit

4. Analog Front End (AFE) & Power Stage

- **DAC Interface** (High-Speed)
- **Reconstruction Low-Pass Filter** (Images Removal)
- **Buffer / Op-Amp** (Gain & Impedance Match)
- **Band-Pass Filter** (100 – 500 kHz)
- **Variable Gain Amplifier** (Controlled by STM32)
- **Power Amplifier (PA)** (High Efficiency, 10W Class-D)
- **Sonar Transducer** (Electrical to Acoustic)

5. Output Testing & Validation Domain

Time-Domain Validation (Oscilloscope)

- Pulse Shape
- Duration
- Amplitude
- Repetition

Frequency-Domain Validation (FFT / Spectrum Analyzer)

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-(2\pi kn/N}$$

- Center Frequency
- Bandwidth
- Sidelobes
- Harmonics

Performance Metrics

- Frequency Accuracy
- Bandwidth Accuracy
- Time Duration Accuracy
- Amplitude Stability
- SNR & Sidelobes

Validate the transmitted acoustic waveform

SYSTEM PIPELINE SUMMARY

Environmental Inputs (Sensors) → STM32G474 Adaptive Control (What to transmit) → PolarFire SoC FPGA Real-Time Engine (How to generate) → DAC + Analog Front End Signal Conditioning & Power Stage → Sonar Transducer Acoustic Wave Generation → Testing & Validation Oscilloscope / FFT Spectrum Analyzer

Problem Solved:
Efficient, adaptive and secure sonar transmissions in complex underwater environments with intelligent control and deterministic real-time waveform generation.

Key Features:

- ✅ Adaptive Waveform Selection
- ✅ Energy-Aware Transmission
- ✅ Deterministic FPGA Engine
- ✅ Low-Power & Efficient
- ✅ Secure & Reliable Operation